



**ARKA JAIN
University**
Jharkhand



Academic Year (AY): 2026-27 (July-December)

Assignment-I

Subject- Biology for Engineer

Programme: BTECH CSE, Semester-7th

Question No.	Text of question	Knowledge Level (KL)
1. (i)	Justify the need to study Biology with reference to environment and technology.	I
(ii)	A device that is used to locate objects in military operations and locate the path of the drill precisely while drilling under seas is called _____. Discuss its bio mimicry.	
(iii)	Write two application of biosensor.	
(iv)	Give two names of engineers with their discovery who won nobel prize in biology and chemistry field? .	
(v)	Why 'Shinkansen' Japan's fastest train is redesign?	
2.	Explain Engineering designs inspired by examples of biology using proper examples.	III
3.	Briefly write about discoveries in the field of biology and medicinal chemistry by engineers.	IV
4.	How Brownian motion in thermodynamic can explain?	II
5.	Define Biosensor. What are the components of biosensor? Explain the working principle of biosensor.	III
6.	Write down about the significance of biology in engineering.	II
7.	Compare the human eye and a camera to highlight differences between biological and engineered systems	II
8.	Distinguish between the Scientific Enquiry and Engineering Method of research.	IV

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Assignment-II

Subject- Biology for Engineer

Programme: BTECH CSE, Semester-7th

Question No.	Text of question	Knowledge Level (KL)
1. (i)	Write the name of buffer used to main the pH of Blood.	I
(ii)	Write the principle of fluorescence spectroscopy.	
(iii)	Write down the component of fluorescence microscope.	
(iv)	Give the name of different stages of sterilization methods.	
(v)	How buffer solution and pH can correlate.	
2.	Illustrate the Role of Buffer Solution in human body giving proper mechanism.	IV
3.	Briefly explain different phases of the bacterial growth curve.	IV
4.	What are Model Organisms? Give brief notes on any three model organisms.	II
5.	Distinguish between biosafety level 1, 2, 3, and 4.	II
6.	why buffer solutions are important in biological systems	IV
7.	How fluorescence microscopy works? Write the principle of fluorescence spectroscopy.Explain electron transition in fluorescence spectroscopy.	III
8.	Discuss about the sterilization various techniques used in Laboratory.	II

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Academic Year (AY): 2026-27 (July-December)

Assignment-III

Subject- Biology for Engineer

Programme: BTECH CSE, Semester-7th

Question No.	Text of question	Knowledge Level (KL)
1. (i)	What do you understand by saprophytic organisms?	I
(ii)	Define taxonomy?	
(iii)	Describe Amminotelsim.	
(iv)	What are the three domines (kingdoms) of life?	
(v)	Draw the ultra structure of Prokaryotic Cell	
2.	Illustrate about hierarchy classification of living organism.	II
3.	Compare the characteristics of Prokaryotic and Eukaryotic cell.	IV
4.	How autotrophs utilize carbon and energy?	I
5.	Explain different mode of excretion in Urioteliic organisms.	II
6.	Classify five kingdom of living being with proper example.	III
7.	Explain the classification of organisms based on carbon utilization of organisms.	II
8.	Differentiate between Unicellular and Multicellular organisms.	IV

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Academic Year (AY): 2026-27 (July-December)

Assignment-IV

Subject- Biology for Engineer

Programme: BTECH CSE, Semester-7th

Question No.	Text of question	Knowledge Level (KL)
1. (i)	Difference between essential and non essential aminoacids.	I
(ii)	Describe the general structure of amino acids.	
(iii)	How are Lipids classified?	
(iv)	There is no life without enzymes. Justify the statement.	
(v)	What are reducing and non-reducing sugars?	
2.	Explain enzyme kinetics and factors affecting enzyme activity.	I
3.	What are lipids? Classify and explain different types of lipids.	II
4.	Biological classification of amino acids and their importance.	IV
5.	Distinguish between RNA and DNA.	IV
6.	Classify proteins based on solubility, shape, and composition with proper diagram.	III
7.	Write a note on: 1. Sugar 2. Starch 3. Cellulose	IV
8.	Explain the mechanism of action of enzymes?	II


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Assignment-V

Subject- Biology for Engineer

Programme: BTECH CSE, Semester-7th

Question No.	Text of question	Knowledge Level (KL)
1. (i)	Draw a neat diagram of DNA double helix structure?	I
(ii)	Write full form of M-RNA & TRNA & their functions?	
(iii)	Give an account on dominant & recessive uses	
(iv)	Define Mendel 1 st & 2 nd law.	
(v)	Define gene mapping?	
2.	What are the three Laws of Inheritance proposed by Mendel? Explain Monohybrid cross.	IV
3.	Elaborate the division in meiosis and mitosis with the help of diagram.	III
4.	State the salient features of the genetic code.	II
5.	Describe Complementary Gene Interaction.	II
6.	Illustrate the importance of the double helix structure of DNA in the context of genetic information storage and transmission.	IV
7.	Explain the roles of mRNA, tRNA, and rRNA in protein synthesis and why each is crucial for this cellular process.	III
8.	Define gene Interaction. Give brief account on Dominant Epistasis with suitable example.	I

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